

## PHARMACOLOGY: PHARMACODYNAMICS

### Practice Quiz

➤ Pharmacology bridges physiology, pathophysiology, and clinical practice.

### *Science of Successful Learning:*

#### Smart Study Tips

**Set specific and *achievable* goals that will focus and track your progress.**

- Practice retrieval: Test yourself regularly.
- Space out your study: Studying material repeatedly over time helps your brain consolidate information and is more effective than cramming.
- Mix up the topics: Switching between subjects and types of problems strengthens memory retrieval and promotes better conceptual understanding.
- Elaborate: Ask yourself **why** and **how** questions to add meaning to your learning and create richer mental connections.
- Dual coding: Draw and label figures, diagrams, flowcharts, etc.

**Get enough sleep and drink enough water.**

### **Goals of Practice Quizzes:**

1. To help students identify what they know and what they understand by applying pharmacology knowledge to clinical case vignettes.
2. To help students get used to thinking through the “Board-style” case vignettes and questions. NYITCOM exams, COMLEX, and USMLE test students’ medical knowledge using these types of questions.

**Note:** Some items may have more than one answer.

**My recommendation** is to home in on the information you need to answer the question, which, at this time, is related to pharmacodynamics. *Don’t let yourself get distracted by the background information that is not relevant for answering the question.*

**CHALLENGE:** Cover the options and try to answer the questions without looking at the options.

### **YOU CAN EASILY FIND THE ANSWERS.**

**“Search” my Notes Handout (“Notes”).** All answers can be found there. (You do not need to read the whole document if you do not wish to.)

If you prefer reading the textbook, links are provided on the PowerPoint and in the CPG.

***Putting in the effort to find the answers cements learning much better than rote memorization.***

Answer Key: Pharmacology concepts are explained in a physiologic and clinical contexts.

**THE ANSWER KEY with explanations will be POSTED Friday November 21.**

### First, the basics.

1. The pharmacokinetics and pharmacodynamics properties of a drug influence the dose-response relationship of that drug. What feature of rational drug dosing links the two properties?

- A. Absorption
- B. Concentration
- C. Intrinsic activity
- D. Potency

Hint: Notes page 2

2. Finish this sentence: A drug with a low  $K_d$  would have high \_\_\_\_\_

- A. Affinity
- B. Efficacy
- C. Functional response
- D. Potency
- E. Specificity

Hint: Notes page 5

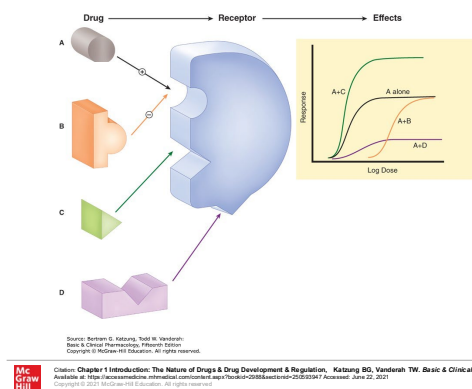
3. Fill in the blank: EC50 is useful for measuring the relative \_\_\_\_\_ of a two or more drugs.

- A. Potency
- B. Affinity
- C. Efficacy
- D. Functional response
- E. Specificity

Hint: Page 7 of Notes Handout

4. Looking at the figure on the right, which one of the following options graphically describes a competitive antagonist?

- A. A alone
- B. A+B
- C. A+C
- D. A+D



### Hint:

The models of the interactions of two drugs at a given receptor is depicted on Part 2, Slide 15 and in the Notes on pages 12 and 13.

The figure above right shows a different way of illustrating the same concept. Try explaining to yourself and your study mates what A, B, C, and D represent and then the effect of each as represented in the graph.

If you want to check yourself, the illustration is explained in the Katzung textbook, Chapter 1, Figure 1-1: <https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3461&sectionid=285588535>

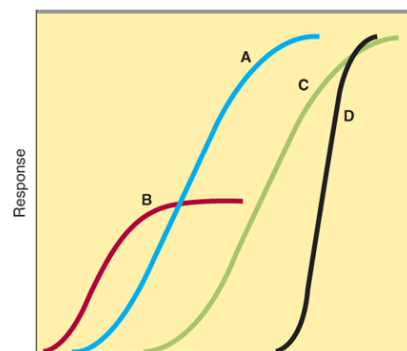
5. An investigational opioid analgesic agent, Drug B, undergoes preclinical testing to determine its potency and efficacy compared to the currently available analgesic drugs, Drugs A, C, D, in accordance with US Food and Drug Administration (FDA) regulations. They all have the same therapeutic mechanism of action. The graded dose-response curves are shown in the graph.

Which drug is the most potent?

Short answer.

Hint: Apply your knowledge of relative EC<sub>50</sub> values to answer this question.

If you want to check yourself, see Notes p8.



Source: Bertram G. Katzung, Todd W. Vanderah:  
Basic & Clinical Pharmacology, Fifteenth Edition  
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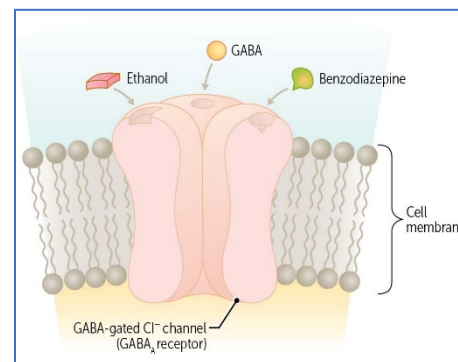
6. Benzodiazepines are positive allosteric modulators at GABA-A receptors. GABA-A receptors are ion channels with 5 subunits and are activated by the neurotransmitter, GABA. GABA binds to its two active sites at the interface between the  $\alpha$  and  $\beta$  subunits, allowing negatively charged  $\text{Cl}^-$  ions to move down its electrochemical gradient into the cell, resulting in hyperpolarization – which leads to inhibition of neuronal firing.

i) What type of receptor is the target of the benzodiazepines?

- A. Ligand-gated ion channel
- B. Nuclear receptor
- C. Receptor tyrosine kinase
- D. Solute carrier
- E. Voltage-gated ion channel

ii) What is the effect of benzodiazepines (simple answer)?

Note: Please try to deduce the answer from the description even though this receptor is not specifically mentioned in the lecture or Notes.



ScholarRx

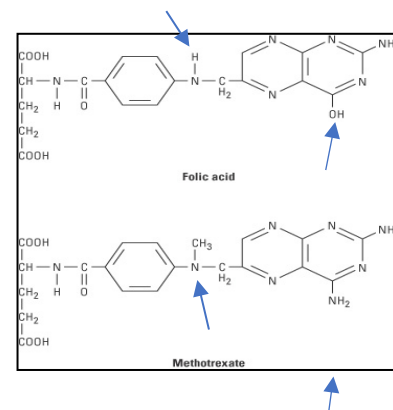
7. A 7-year-old child's treatment for acute lymphoblastic leukemia includes methotrexate (lower structure on the figure), which blocks the activation of folic acid (upper molecule) in the cancer cells. This action leads to cancer cell apoptosis. Methotrexate is a structural analog of folic acid.

The arrows are pointing to the differences in the molecules.

This description is an example of what pharmacologic concept?

Short answer

Hint: Think about the pharmacologic concept exemplified by the figure.



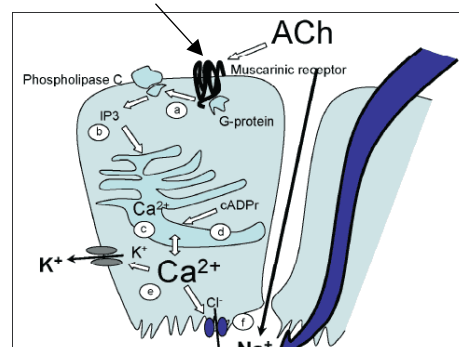
8. A 57-year-old woman with primary Sjogren's syndrome complains that the saliva substitute the nurse practitioner suggested isn't helping with her dry mouth. She agrees to take oral pilocarpine, which induces saliva secretion by binding to the muscarinic M3 receptors on the surface of secretory acini cells of salivary glands. The drug's effect is equivalent to that produced by the natural ligand of M3 receptors, acetylcholine (ACh).

i) What is the mechanism of action of the prescribed drug?

One sentence

ii) Draw a graphic representation and label it.

Definition: Sjogren's syndrome is a slowly progressing, chronic autoimmune disease affecting exocrine glands.



Muscarinic Receptor Types in the Alimentary Canal

[https://www.jpp.krakow.pl/journal/archive/0309/articles/01\\_article.html](https://www.jpp.krakow.pl/journal/archive/0309/articles/01_article.html)

9. A drug development company ran a preclinical trial to evaluate an experimental antiarrhythmic medicine among male and female mouse models of atrial fibrillation. Endpoints were improved functional status (efficacy) and onset of arrhythmia (toxicity). The percentage of individuals that showed the defined effect at each dose level was recorded.

(This is a fictional scenario to my knowledge.)

Which of the following curves would BEST describe the findings in this study?

- A. Drug concentration-time plot
- B. Receptor binding-concentration curve
- C. Graded dose-response curve
- D. Lineweaver-Burk plot
- E. Quantal dose-response curve

10. What is the clinical application of plotting the curves described in the above vignette?

- A. Determining the maximal effect of the drug
- B. Determining the elimination rate of the drug
- C. Determining the relative potency of the drug
- D. Determining the effective dose range of the drug
- E. Determining the safe dose range of the drug

11. A hypotensive patient in the ICU is given an intravenous infusion of an agonist called phenylephrine, which mimics the actions of the natural ligand, epinephrine. The drug acts on receptors in the blood vessels, causing vasoconstriction, which increases blood pressure. The drug is a small molecule chemically similar to the natural ligand and produces the maximal effect.

Which of the following properties ultimately determines phenylephrine's pharmacologic effect (the change in biologic action)?

- A. Hydrophobicity
- B. Potency
- C. Selectivity for vascular receptors
- D. Structure-activity relationship

(Select the one BEST answer.)

12. A 48-year-old man presents to the family practice clinic for pharmacologic management of hypertension and hypercholesterolemia, which have not adequately responded to diet and exercise. The patient has a family history of cardiovascular disease – his father died of a heart attack at age 63 and his mother died of a stroke at age 72. In addition to blood pressure medication, the patient agrees to therapy with a drug that reversibly binds to the active (catalytic) site of a specific enzyme and blocks the hepatic synthesis of cholesterol.

i) What is the mechanism of action of the prescribed drug?

ii) Draw the graph that describes the mechanism of the action experimentally and label it.

Hint #1: Notes page 12.

Hint #2: Highlight the clues that relate to pharmacodynamics concepts. Please don't become distracted by the other information.

13. A 23-year-old medical student spending the summer in a malaria-endemic region takes an antimalarial drug for the prophylaxis (prevention) of malaria. The drug is 99% bound to plasma proteins. What is the change in function that can be expected from this drug-protein interaction?  
Short answer.

14. A 34-year-old female with generalized anxiety disorder is prescribed Xanax (alprazolam). The drug acts by binding an allosteric site (BZ receptor) on GABA-A receptor, which is an ion channel. The drug enhances the receptor response to the natural ligand, GABA, which induces antianxiety and sedative effects.

- i) What is the mechanism of action of the prescribed drug?
- ii) Draw the graph that describes the mechanism of the action experimentally and label it.

15. A study on isolated, perfused animal hearts showed that pindolol caused an increased heart rate, but to a lesser degree than that caused by the natural ligand. However, in the presence of a highly effective stimulant (agonist) acting on the same receptors, pindolol caused a dose-dependent reversible decrease in heart rate.

- i) What is pindolol's mechanism of action (classical receptor occupancy model)?
- ii) Draw the graded dose-response graph that describes the mechanism of the action experimentally and label it.

16. In the two-state receptor occupancy model:

- i) Which one of the following options would best describe pindolol's action?

- A.  $R_a + D_a$
- B.  $R_a + D_{pa}$
- C.  $R_a + D_{ant} + R_i + D_{ant}$
- D.  $R_i + D_i$

- ii) Draw the graph that describes each of the options and label it.

17. A 79-year-old man is hospitalized for the treatment of decompensated heart failure. Continuous intravenous infusion of an inotrope is administered. The drug activates beta adrenergic receptors (GPCRs) in the heart, which increases strength of contraction, which leads to increased stroke volume and cardiac output. An effect of this drug is a reduction in response with continuous use of the drug due to endocytosis of the beta receptors.

What is the term used for this response to the continuous use of the inotrope in this scenario?

Short answer.

Hint: Notes p16

18. The mother of an unconscious 4-year-old boy with a one-year-history of type 1 diabetes mellitus treated with insulin administers a subcutaneous injection of glucagon to reverse the hypoglycemia caused by an overdose of insulin.

Which of the following best describes the action of glucagon in the context of the pharmacology?

- A. Full agonist
- B. Partial agonist
- C. Competitive antagonist
- D. Noncompetitive antagonist
- E. Physiological antagonist
- F. Chemical antagonist

Definitions:

Insulin is a hormone secreted by the endocrine pancreas; it stimulates cellular uptake of glucose, which reduces blood glucose concentrations.

Glucagon is a hormone secreted by the endocrine pancreas that stimulates glucose formation in the liver, which increases blood glucose.

Insulin and glucagon are both formulated for pharmacologic use.

19. A 15-year-old male is prescribed oral tetracycline by the dermatologist for the treatment of acne. He is cautioned not to take the medication together with an antacid, dairy products, or vitamins with minerals. He should separate taking the antibiotic and the other products by at least 2 hours before or after taking the acne medication.

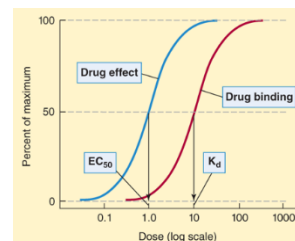
What mechanism of antagonism is described?

Short answer.

20. An experimental agonist molecule tested in animal studies against an irreversible antagonist shows that its  $EC_{50}$  is  $1/10^{\text{th}}$  (0.1%) of its  $K_d$ .

Which one of the following options best describes this finding?

- A. Binding to inert molecules
- B. Noncompetitive antagonism
- C. Partial agonism
- D. Spare receptors



21. Sixteen patients with epidermoid carcinomas of the head and neck received maximally tolerated doses of methotrexate alone as 30-hr intravenous infusions at 2-week intervals. 44% of patients responded. Significant bone marrow suppression and 4 deaths were incurred. The cumulative distribution curve to the right shows that the drug has a low \_\_\_\_\_.  
Fill in the answer.

